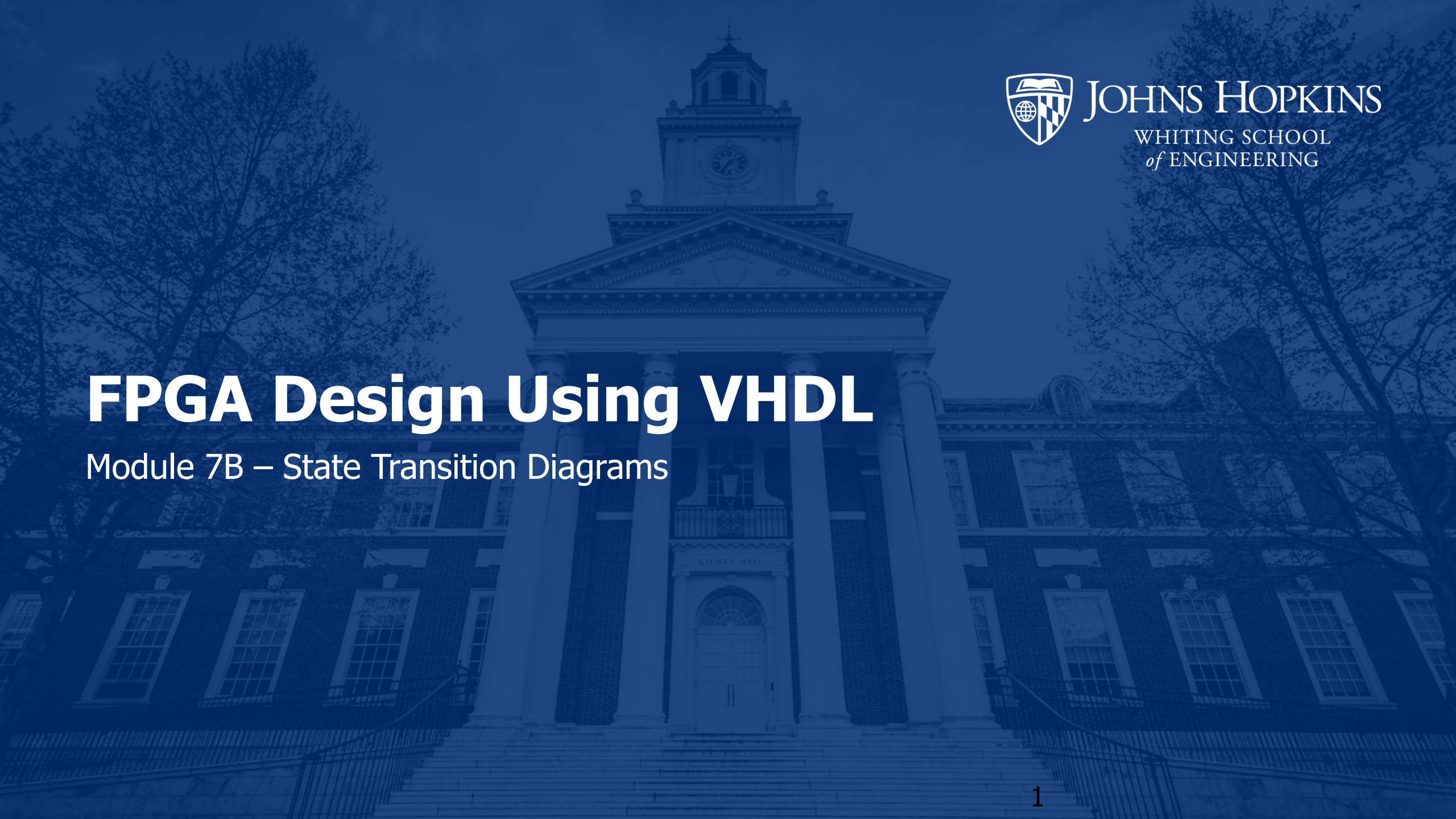
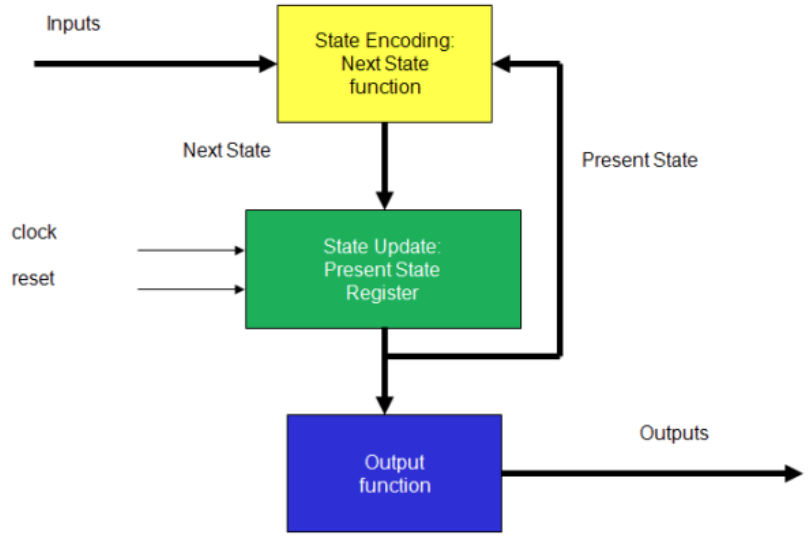


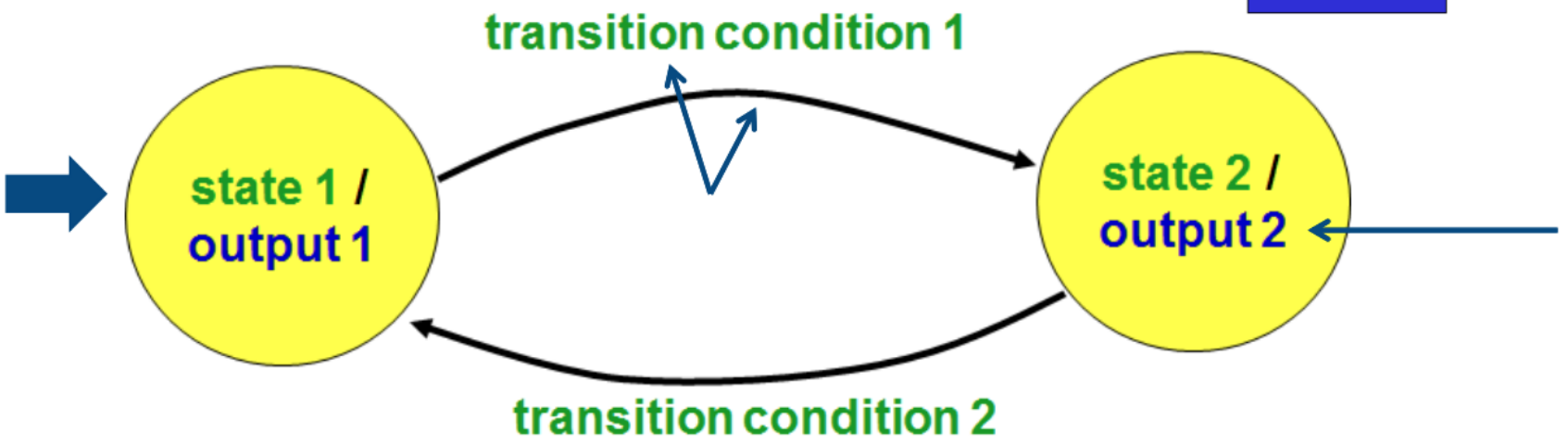
FPGA Design Using VHDL

Module 7B – State Transition Diagrams

Moore FSM

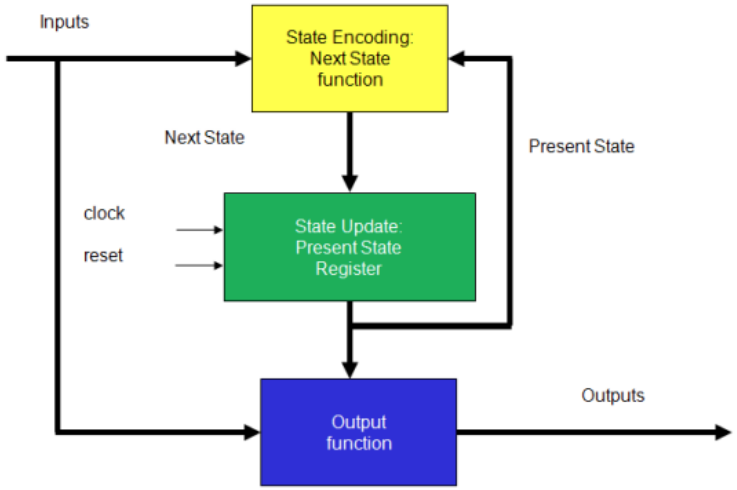
State Transition Diagram

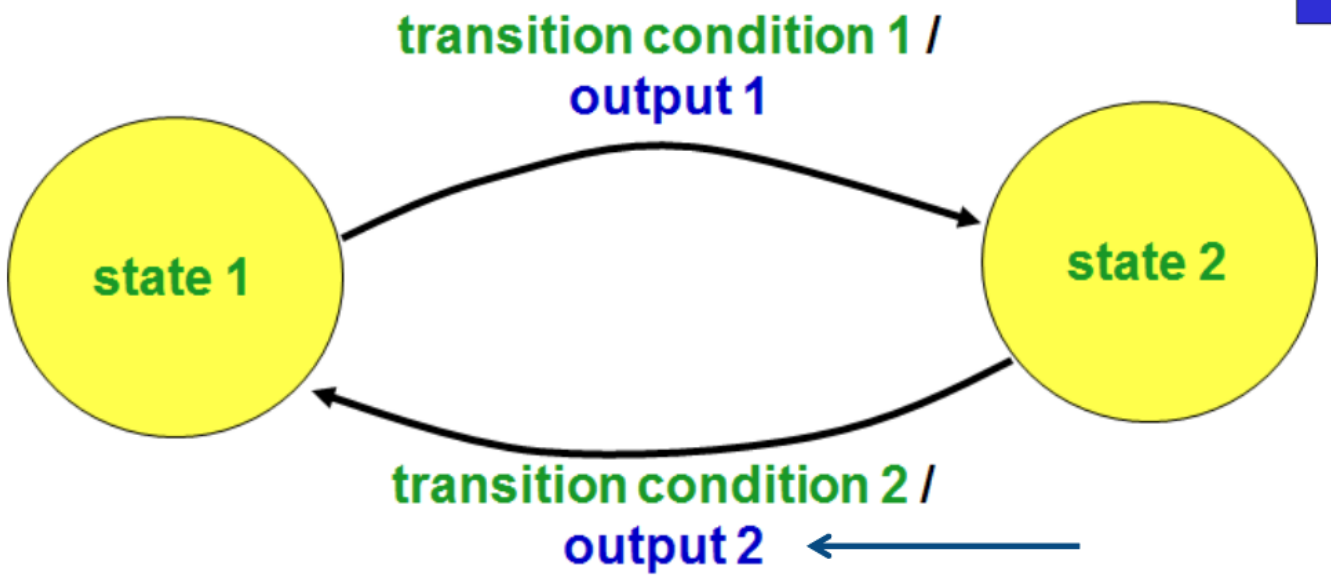




Mealy FSM

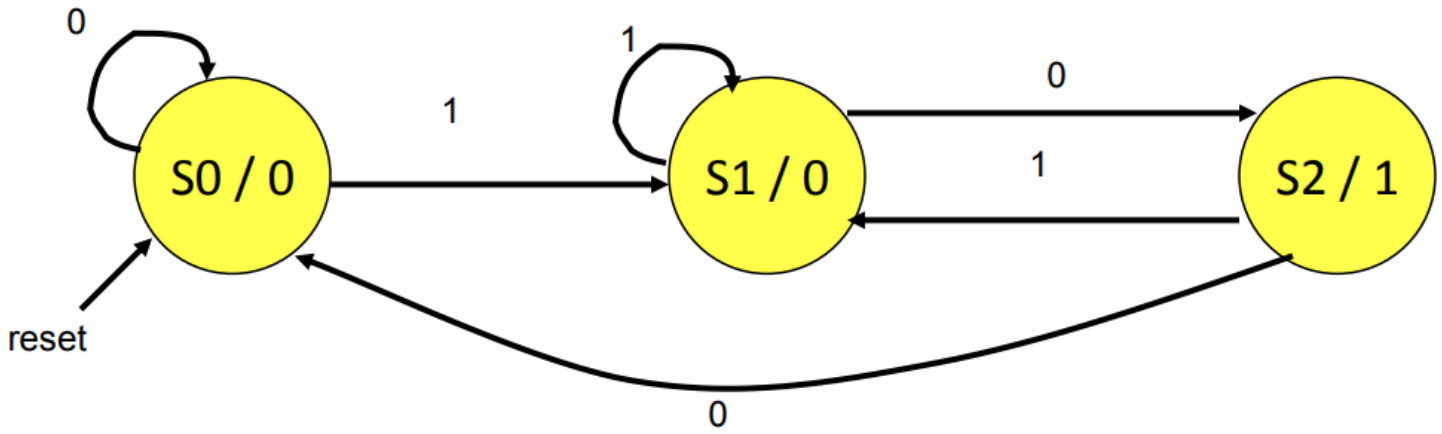
State Transition Diagram





Moore FSM Example

- Moore FSM that recognizes sequence "10"
- Set output Y to 1 when sequence "10" is detected
- Example: 0000101011100001101011110000



Meaning of states:

S0: No elements of the sequence observed

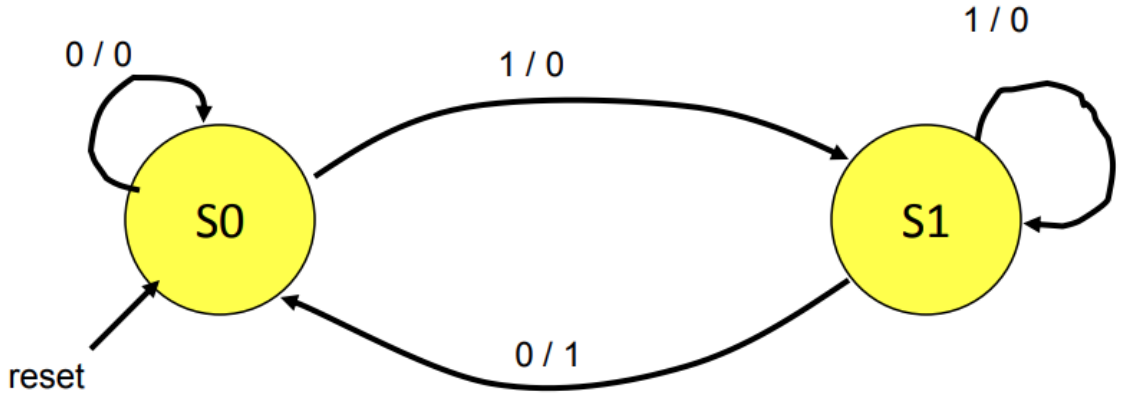
S1: "1" observed

S2: "10" observed

Sequence detectors will be on quiz and exam

Mealy FSM Example

- Mealy FSM that recognizes sequence "10"\
- Set output Y to 1 when sequence "10" is deected
- Example: 0000101011100001101011110000



Meaning of states:

S0: No elements of the sequence observed

S1: "1" observed